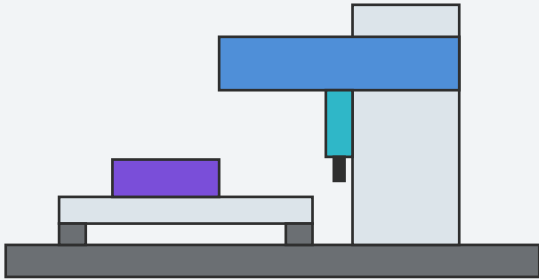


Machines in a Manufacturing Facility

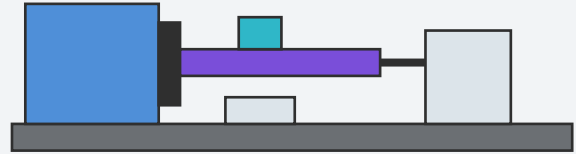
Simplified schematic drawings with typical failure modes and the signals used for predictive maintenance



1. CNC milling machine

Typical failure: Spindle bearing wear, cutter wear

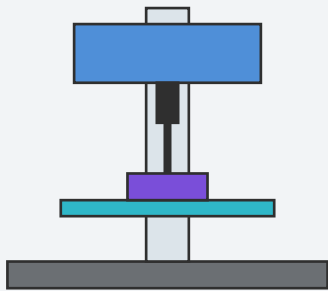
Monitor: Vibration, spindle current, sound



2. CNC lathe

Typical failure: Headstock bearing wear, chuck run-out

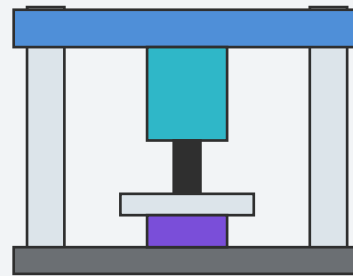
Monitor: Vibration, motor current, temperature



3. Drilling machine

Typical failure: Spindle bearing and belt wear

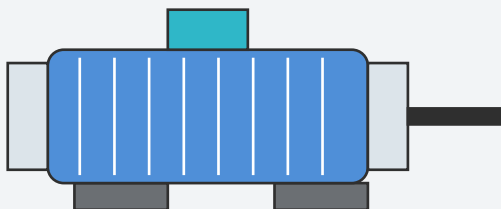
Monitor: Vibration, sound



4. Hydraulic press

Typical failure: Seal leaks, pump wear, overheating

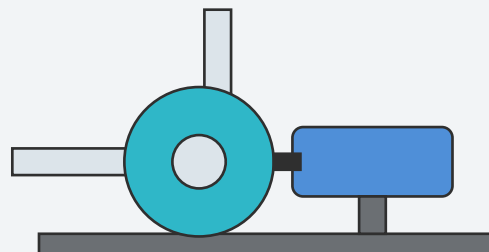
Monitor: Pressure, oil temperature, vibration



5. Electric motor

Typical failure: Bearing fault, winding overheating, imbalance

Monitor: Vibration, current, temperature



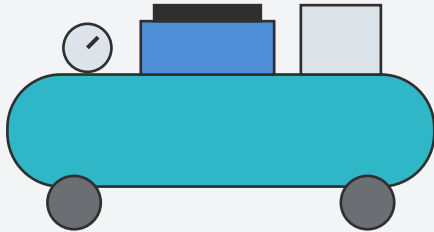
6. Centrifugal / coolant pump

Typical failure: Bearing wear, cavitation, seal leak

Monitor: Vibration, sound, flow, current

Machines in a Manufacturing Facility

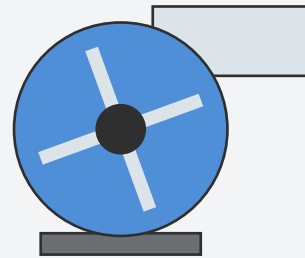
Simplified schematic drawings with typical failure modes and the signals used for predictive maintenance



7. Air compressor

Typical failure: Valve wear, bearing failure, overheating

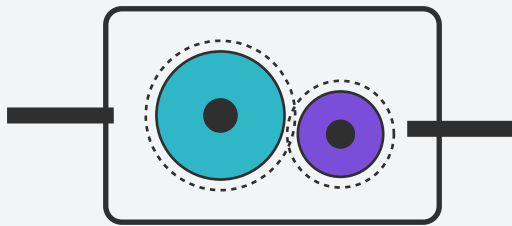
Monitor: Vibration, temperature, pressure, current



8. Fan / blower

Typical failure: Imbalance, bearing wear, blade damage

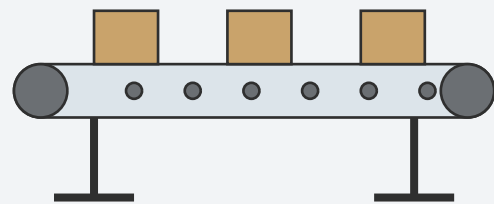
Monitor: Vibration, sound, current



9. Gearbox / reducer

Typical failure: Gear tooth wear, bearing fault, oil breakdown

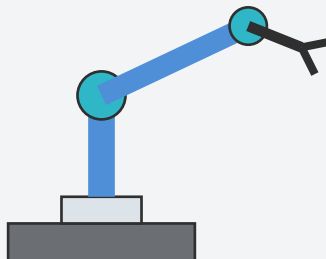
Monitor: Vibration, oil temperature, sound



10. Belt conveyor

Typical failure: Roller bearing seizure, belt slip, motor overload

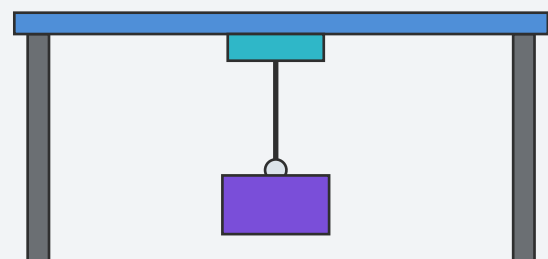
Monitor: Motor current, vibration, temperature



11. Industrial robotic arm

Typical failure: Joint gearbox wear, motor overheating

Monitor: Joint torque, vibration, temperature



12. Overhead crane / hoist

Typical failure: Brake wear, hoist motor and rope wear

Monitor: Motor current, vibration, load

Machines in a Manufacturing Facility

Simplified schematic drawings with typical failure modes and the signals used for predictive maintenance

Other machines found in a facility

Machine tools

Grinding machines, boring and broaching machines, shaping and planing machines, EDM and wire-cut machines

Forming and fabrication

Mechanical presses, forging hammers, press brakes, shearing machines, laser and plasma cutters, injection moulding and die-casting machines, welding machines and robots

Rotating equipment

Turbines, generators, additional gearboxes and reducers

Material handling

Forklifts, automated guided vehicles (AGVs), hoists, palletising robots

Utilities and support

Boilers, furnaces, heat treatment ovens, chillers, cooling towers, HVAC, transformers, DG sets, UPS units, hydraulic power packs, dust collectors

Assembly, inspection and packaging

Assembly lines and fixtures, coordinate measuring machines (CMM), leak, pressure and torque testing rigs, packaging, labelling and filling machines

Best targets for the predictive maintenance problem

Electric motors, pumps, compressors, fans and gearboxes (items 5, 6, 7, 8 and 9) share the same bearing-driven failure pattern, so one method can cover them all. The CNC spindle (item 1) is the most relevant to Mysuru machining units.